

Fundamentals of Solid State Physics

Semiconductors - Carrier Behaviors

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This Class

- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- **Electronic Properties (Week 4–12)**
 - Free electrons (the Drude and Sommerfeld models)
 - Electrons in a periodic potential (Bloch's Theorem)
 - The near-free electron model, the tight-binding model
 - Electronic band diagram, band gaps, effective mass
 - Metals, insulators, semiconductors
 - Devices: junctions, diodes, transistors
- Thermal Properties (Week 13)
- Optical Properties (Week 14)
- Magnetic Properties (Week 15)

Carrier Behaviors in Semiconductors

- Equilibrium Carriers (热平衡态载流子)
- Non-equilibrium Carriers (非平衡态载流子)
- Majority & Minority Carriers 多数 / 少数 载流子
- Current Flow
 - drift current
 - diffusion current
- Generation 产生
- Recombination 复合

Equilibrium carriers (热平衡态载流子)

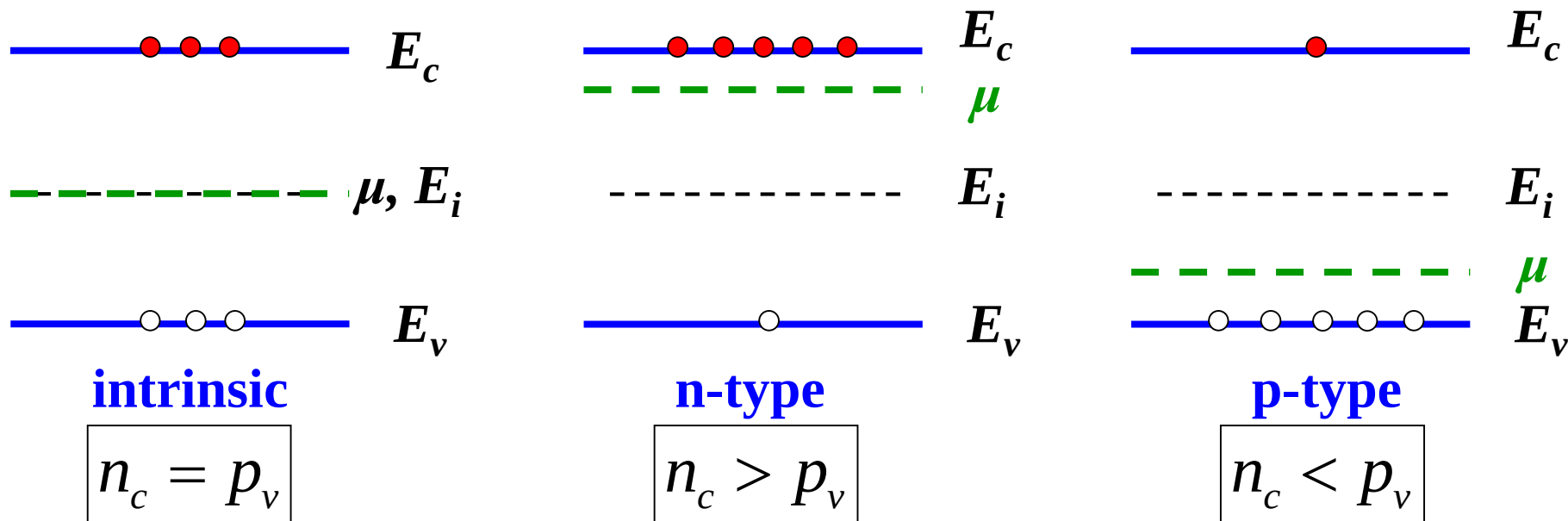
only thermal activation

$$n_c = N_c(T) e^{-(E_c - \mu)/k_B T}$$

$$p_v = P_v(T) e^{-(\mu - E_v)/k_B T}$$

$$n_c p_v = N_c(T) P_v(T) e^{-E_g/k_B T} = n_i^2$$

mass action law



Non-equilibrium carriers (非平衡态载流子)

activation by other energy sources
e.g., photon absorption, current injection, ...

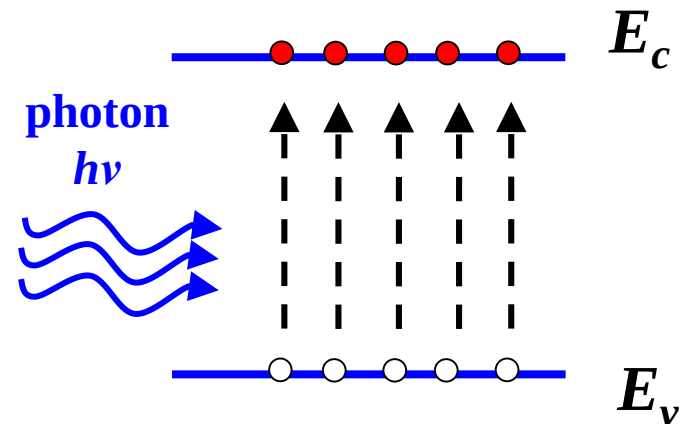
extra more carriers n_c and p_v

$$p_v \neq P_v(T)e^{-(\mu-E_v)/k_B T}$$

$$n_c \neq N_c(T)e^{-(E_c-\mu)/k_B T}$$

$$n_c p_v \neq N_c(T)P_v(T)e^{-E_g/k_B T} = n_i^2$$

mass action law is not valid



Non-equilibrium carriers (非平衡态载流子)

activation by other energy sources

e.g., photon absorption, current injection, ...

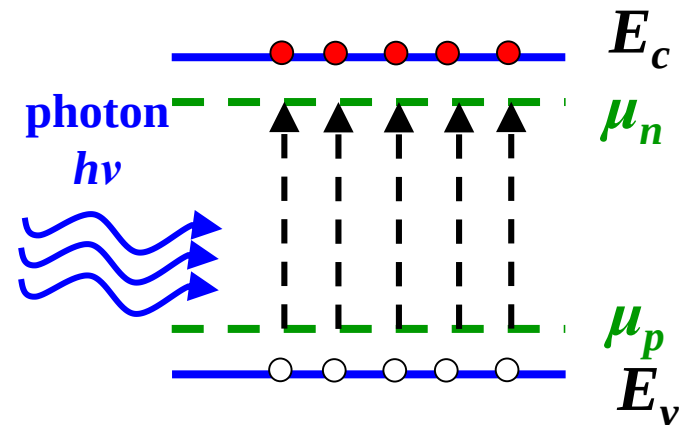
extra more carriers n_c and p_v

rewrite:

$$p_v = P_v(T) e^{-(\mu_p - E_v)/k_B T}$$

$$n_c = N_c(T) e^{-(E_c - \mu_n)/k_B T}$$

$$\mu_n \neq \mu_p$$



μ_n, μ_p : quasi-Fermi levels (准费米能级)

Non-equilibrium carriers (非平衡态载流子)

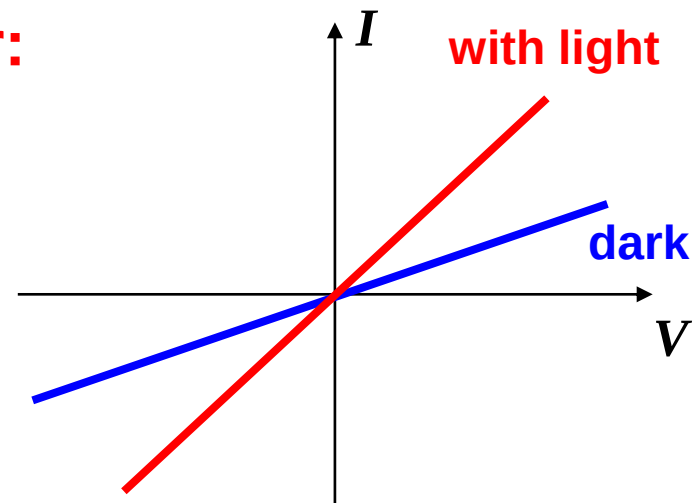
activation by other energy sources
e.g., photon absorption, current injection, ...

extra more carriers n_c and p_v

light absorption increases the
conductivity of a semiconductor:

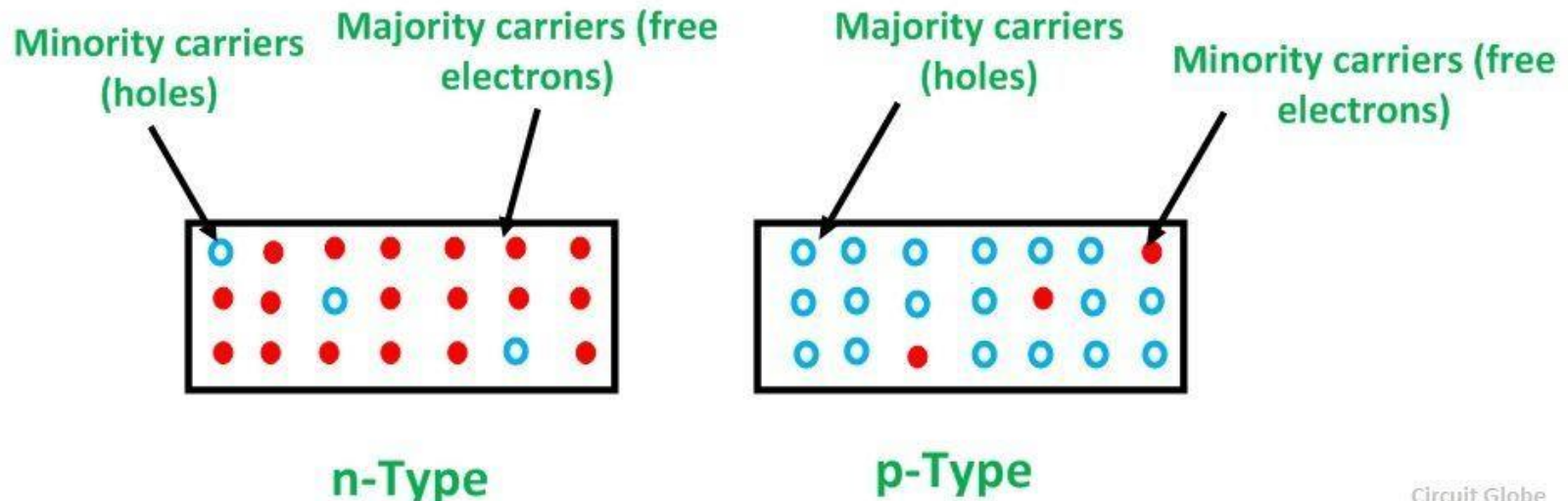
---- photoconductor

$$\sigma = n_c e \mu_e + p_v e \mu_h$$



Majority & Minority Carriers

- Majority carriers (多数载流子)
- Minority carriers (少数载流子)
 - minority carriers are very important, because their amount can be easily changed by current injection, optical absorption, etc.

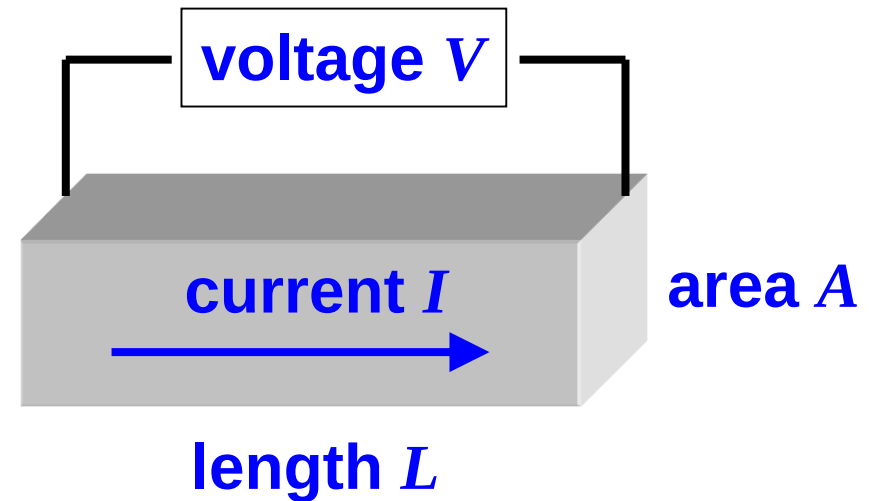


Current Flow

- Drift current 迁移电流
 - caused by electric field
- Diffusion current 扩散电流
 - caused by concentration gradient

Current Flow

- Drift current 迁移电流
 - ▣ caused by electric field



Ohm's Law

$$j = \sigma E = nq\mu E = nqv$$

n - carrier concentration (#/m³)
 D - diffusivity (m²/s) 扩散系数
 μ - mobility (m²/V/s)

Current Flow

- Diffusion current 扩散电流
 - caused by concentration gradient

ink in water



1D

$$j = -qD \frac{\partial n}{\partial x}$$

3D

$$\mathbf{j} = -q\mathbf{D}\nabla n$$

diffusivity 扩散系数

$$D = \mu \frac{kT}{q}$$

n - carrier concentration ($\#/m^3$)

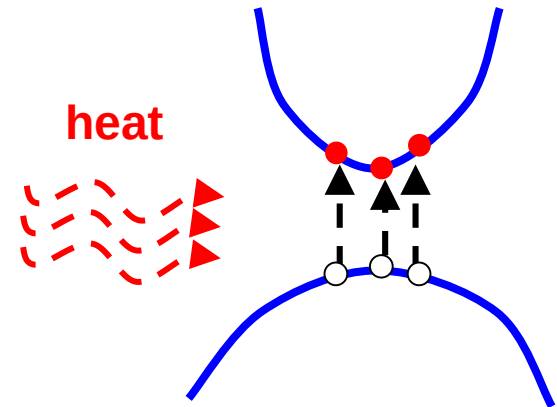
D - diffusivity (m^2/s) 扩散系数

μ - mobility ($m^2/V/s$)

Carrier Generation

- Excited by thermal energy

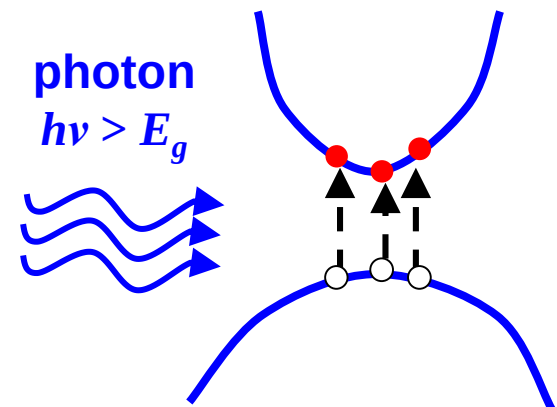
$$n_c p_v = N_c(T) P_v(T) e^{-E_g/k_B T} = n_i^2$$



- Excited by photons

- photodetector
- solar cells
- ...

$$h\nu = e + h$$



- Other energy sources

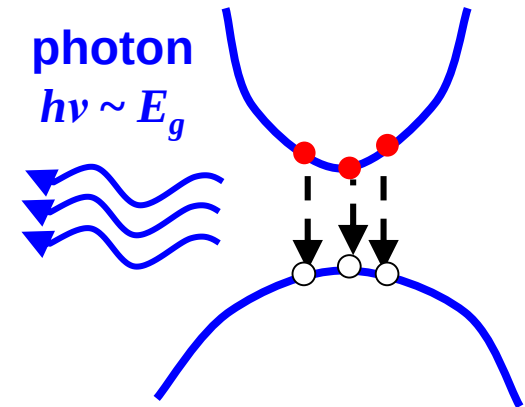
- ...

Carrier Recombination 复合

■ Radiative Recombination 辐射复合

- ❑ LED
- ❑ laser
- ❑ ...

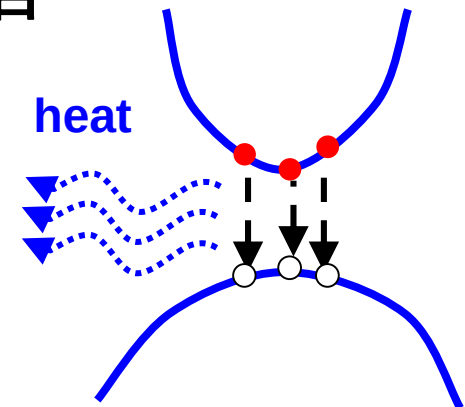
$$\mathbf{e} + \mathbf{h} = \text{photon}$$



■ Non-radiative recombination 非辐射复合

- ❑ defects, impurities, surfaces
- ❑ lattice (phonons)
- ❑ Auger process
- ❑ ...

$$\mathbf{e} + \mathbf{h} = \text{phonons}$$

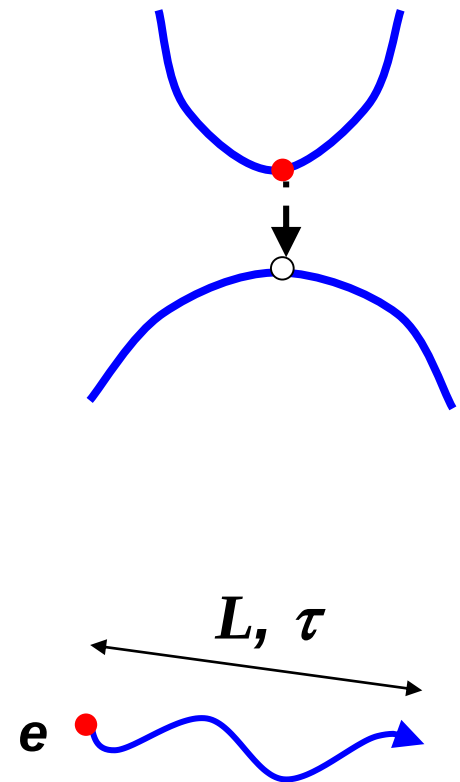


Carrier Recombination 复合

- 载流子寿命 Carrier lifetime τ_n and τ_p
 - the averaged time before the non-equilibrium carriers recombine
- 扩散系数 Diffusivity D_n and D_p
- 扩散长度 Diffusion length L_n and L_p
 - the averaged distance carriers move before recombination

$$L = \sqrt{D\tau}$$

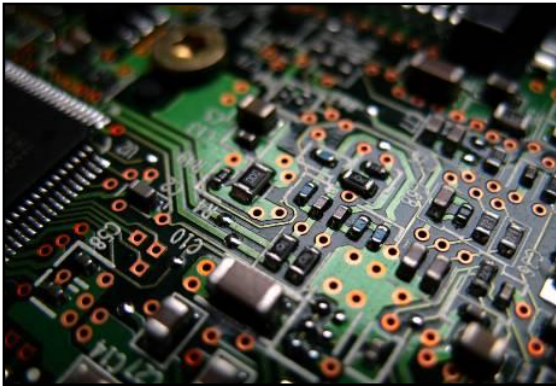
- At non-equilibrium, **minority carrier** diffusion and lifetime are important for device performance



D - diffusivity (m^2/s) 扩散系数

Semiconductors - Applications

different carrier behaviors offer different applications



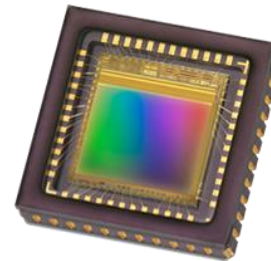
integrated circuits



LEDs



lasers



detectors



solar cells

Thank you for your attention